Error-Aware Nonlinear Gain-Based PID Tracking Control: A Constructive and Self-Tuning Approach

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***Abstract*—Conventional proportional–integral–derivative (PID) controllers face fundamental limitations in handling unknown nonlinearities and coupling effects, often requiring manual tuning or complex adaptation mechanisms. This article introduces a novel nonlinear PID tracking control architecture while incorporating three innovative features: first, analytically constructed error-aware gains with self-tuning capability, utilizing current, past, and predicted error information; second, inherent antiwindup properties through nonlinear integral design; and third, fully autonomous operation without the need for precise a priori knowledge of system dynamics or manual intervention. Unlike prior works mainly restricted to single-input-single-output systems and only achieving uniformly ultimately bounded stability, our approach handles higher order multiple-input–multiple-output systems with unknown coupled dynamics, achieving asymptotic tracking. The controller maintains structural PID simplicity while providing enhanced nonlinear compensation and robust adaptive performance. Comprehensive Lyapunov analysis rigorously validates stability for general normal form nonlinear systems, with superior performance in both regulation and time-varying reference tracking scenarios.**

***Index Terms*—Adaptive nonlinear proportional–integral–derivative (PID), error-aware gain, nonlinear systems, tracking control.**

I. Introduction

Proportional–integral–derivative (PID) controllers, renowned for their simple structure and ease of implementation, along with their comprehensive regulation mechanism that integrates the "present" (proportional action)–the "past" (integral action)–the "tendency" (derivative action), have become one of the most widely applied control strategies [1], [2], [3]. For nonlinear systems, how to achieve stable and reliable control through well-designed PID gains has emerged as a research hotspot in the field. Establishing an effective PID gain-tuning mechanism remains an unresolved challenge.

Most existing PID control methods can generally be categorized into two types: robust linear PID control with constant gains [4], [5], [6] and adaptive linear PID control with time-varying gains [7], [8], [9]. In the former, the gains remain fixed once determined, while the latter dynamically adjusts the gains through real-time updates of one or more adaptive parameters. It should be noted that fixed linear PID control methods based on constant gains exhibit significant performance limitations when dealing with system nonlinearities, uncertainties, and complex dynamic environmental changes. These limitations primarily stem from the inherent linear structure of PID controllers, which struggle to effectively capture nonlinear characteristics, resulting in insufficient flexibility and adaptability. Consequently, control performance becomes unpredictable, and stability is difficult to guarantee. In addition, the inherent integral windup phenomenon is hard to avoid, often leading to delayed system response or even instability [10], [11]. Although introducing time-varying gains can enhance adaptability to some extent, the control structure remains fundamentally linear, making it difficult to fundamentally overcome performance degradation caused by nonlinearities. Furthermore, most current PID controller gain designs still rely heavily on "trial-and-error" or empirical rules, lacking systematic and theoretical guidance, which complicates the parameter-tuning process and makes it challenging to ensure robustness and stability [5].

To address these issues, the authors in [12] and [13], by combining robust adaptive control theory with Lyapunov stability analysis, proposed an online auto-tuning strategy for PID gains, reducing reliance on manual parameter adjustments. Building on this, Xiang et al.'s team [14] further introduced a nonlinear PID control method by incorporating nonlinear combinations of PID elements, significantly improving the controller's performance in handling nonlinear and uncertain systems. Moreover, to validate the effectiveness and theoretical foundation of nonlinear PID controllers in practical systems, Zhao et al. [15] systematically analyzed the conditions under which second-order nonlinear systems fail to achieve stable control with constant-gain linear PID. Then, they proposed a state-aware time-varying gain PID method, achieving global stabilization and set-point asymptotic regulation in closed-loop second-order single-input–single-output (SISO) systems.

Notably, although existing time-varying gain PID control methods can improve system performance to some degree, they still heavily rely on adjustments of single or multiple parameters, leaving integral windup and nonlinearity handling issues unresolved. Therefore, while preserving the intuitiveness and ease of implementation of PID controllers, systematically eliminating integral windup and further enhancing adaptability and robustness against system nonlinearities and uncertainties remain critical challenges in the field of control.

To tackle these challenges, this article proposes a nonlinear adaptive PID control method based on error-aware gain. While inheriting the intuitive structure and ease of implementation of traditional PID controllers, this method endows the controller with inherent nonlinear compensation capabilities, along with adaptive and robust features. Specifically, compared to existing approaches, the proposed method offers the following key advantages.

1) *Error-aware gain regulation:* A gain-tuning mechanism is introduced where the three PID gains are analytically constructed based

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on current, past, and predicted information, and anticipated tracking errors that, respectively, are integrated into the scheme. This establishes an error-aware gain-tuning PID control method where control intensity dynamically adjusts in response to error.

2) *Antiwindup capability:* Unlike traditional fixed-gain linear PID controllers, which are prone to integral windup, the proposed nonlinear PID controller incorporates an error-driven nonlinear integral gain, effectively preventing integral saturation and mitigating instability caused by integral accumulation, thereby improving system stability and reliability.
3) *Autonomous operation and responsiveness:* By directly linking PID gains to system error analytically and incorporating a single parameter tuning mechanism, the control process operates autonomously without manual intervention while enhancing response capabilities.

II. Problem Setting

A. System Description

The multiple-input multiple-output (MIMO) nonlinear dynamic system is described by

$$\begin{cases} \dot{x}_i = x_{i+1}, \quad i = 1, 2, \ldots, n-1 \\ \dot{x}_n = B(x,t)u + F(x,t) + d(t) \\ y = x_1 \end{cases} \tag{1}$$

where $x_i \in \mathbb{R}^m$ $(i = 1, \ldots, n)$ are the substate vectors of the system. $x \in \mathbb{R}^{mn}$ is the full state vector of the system, $B(x,t) \in \mathbb{R}^{m\times m}$ is the time-varying yet uncertain control gain matrix, $u \in \mathbb{R}^m$ is the system control input, and $F(x,t) \in \mathbb{R}^m$ represents the unknown nonlinear function vector. $d(t) \in \mathbb{R}^m$ denotes external disturbances with $\|d(t)\| \le \bar{d}$, where $\bar{d} > 0$ is an unknown positive constant and $y \in \mathbb{R}^m$ is the measurable system output.

We define the output tracking error as: $e = y - y^*$, where $y^* \in \mathbb{R}^m$ represents the desired trajectory. To proceed with the control design, we require the following assumptions.

Assumption 1 (See [16]): The desired tracking trajectories y_j^* $(j = 1, \ldots, m)$ and their derivatives up to the nth-order are known, smooth functions of time and bounded. In addition, the state vector x is available for control design.

Assumption 2 (See [17]): Let $B_p(x,t) = B(x,t) + B^\top(x,t)$ be either positive definite or negative definite, where $B_p(x,t) \in \mathbb{R}^{m\times m}$ is a matrix function.

Assumption 3 (See [18]): Certain crude structural information on $F(x,t)$ is available to allow unknown constant b^* and known scalar-valued smooth function $\psi^*(x)$ to be extracted, such that $\|F(x,t)\| \le b^*\psi^*(x)$ for $\forall x, t$ where $\psi^*(x)$ is bounded if x is bounded.

Remark 1: Assumption 1 is quite standard for tracking control of nonlinear systems (see [16], [20], and the references therein). Assumption 2 is required to ensure the controllability of the system, while Assumption 3 is introduced to extract core structural information from the unknown function $F(x,t)$. In practical applications, the "core function" (which may not be unique) can be readily constructed by leveraging crude structural information about $F(x,t)$ in [12] and [18].

For later technical development, we introduce a filtered (generalized) error signal z as follows:

$$z = \iota_0 e + \iota_1 \dot{e} + \cdots + \iota_{n-3} e^{(n-3)} + e^{(n-2)} \tag{2}$$

where ι_0 to ι_{n-3} are positive constants chosen such that the roots of the polynomial $\theta^{(n-2)} + \iota_{n-3}\theta^{(n-3)} + \cdots + \iota_0 = 0$ have negative real parts. From (2), it can be shown that the convergence of z and $\dot{z}$ guarantees the convergence of e and its derivative up to $(n-1)$th order [19]. Our design will be focused on developing a nonmodel-based control scheme to ensure asymptotic tracking of e,..., $e^{(n-1)}$, which is achieved indirectly by regulating z to zero asymptotically. To this end, we use (1) and (2) to get

$$\begin{aligned} \ddot{z} &= x^{(n)} - y^{*(n)} + \iota_{n-3} e^{(n-1)} + \cdots + \iota_0 \ddot{e} \\ &= B(x,t)u + L(\cdot) \end{aligned} \tag{3}$$

where $L(\cdot) = F(x,t) + d(t) - y^{*(n)} + \iota_{n-3} e^{(n-1)} + \cdots + \iota_0 \ddot{e}$ represents lumped uncertainty.

B. Control Objective

This article aims to develop an error-aware nonlinear gain-based adaptive PID tracking control for the coupled MIMO nonlinear systems (1) such that the output tracking error e converges to zero asymptotically in the presence of uncertain $B(x,t)$, $F(x,t)$, and $d(t)$.

III. Motivation for Nonlinear PID Control

To better understand the motivation behind our design, let us consider the following simple example:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = u + f(x_1, x_2) \end{cases} \tag{4}$$

where $x_1, x_2 \in \mathbb{R}$ are the system states, $u \in \mathbb{R}$ is control input, and $f(x) = x_1^3 + x_2^3 \in \mathbb{R}$ is a nonlinear function. For simplicity, we consider stabilization problem, i.e., design u so that x_1 and x_2 are driven to a small region or zero eventually.

A. Linear PID and Its Limitations

If the conventional linear PID controller is implemented

$$u = -k_p x_1 - k_i \int_0^t x_1(\tau)d\tau - k_d \dot{x}_1 \tag{5}$$

where k_p, k_i, and k_d are appropriate positive constants, then the closed-loop dynamics become

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k_i & -k_p & -k_d \end{bmatrix}}_{\text{Linear subsystem}} \mathbf{x} + \underbrace{\begin{bmatrix} 0 \\ x_1^3 \\ x_2^3 \end{bmatrix}}_{\text{Nonlinearities}} \tag{6}$$

where $\mathbf{x} = [\int_0^t x_1(\tau)d\tau, x_1, \dot{x}_1]^\top \in \mathbb{R}^3$, from which it is observed that although the linear subsystem can be made stable by choosing appropriate k_p, k_i, and k_d, the influence of x_1^3 and x_2^3 on the system remains and cannot be eliminated by such linear PID control, no matter how the PID gains are selected.

The preceding analysis reveals that conventional fixed-gain linear PID controllers fundamentally fail to stabilize the underlying nonlinear system. This inherent limitation motivates a key research inquiry: whether an effective control strategy can be designed under the nonlinear control paradigm. Aiming to tackle this challenging problem, this work explores its feasibility and provides a viable solution in the subsequent sections.

B. Nonlinear PID Control Solution

We consider the following nonlinear gain-based PID control:

$$\begin{aligned} u = &-K_P(x_1, k_p)x_1 - K_I\left(\int_0^t x_1(\tau)d\tau, k_i\right)\int_0^t x_1(\tau)d\tau \\ &- K_D(\dot{x}_1, k_d)\dot{x}_1 \end{aligned} \tag{7}$$

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where $K_P(x_1, k_p)$, $K_I(\int_0^t x_1(\tau)d\tau, k_i)$, and $K_D(\dot{x}_1, k_d)$ are state-aware gains, designed as $k_p + x_1^2$, k_i, and $k_d + \dot{x}_1^2$ with k_p, k_i, and $k_d > 0$, respectively, then the closed-loop system dynamics become

$$\dot{x}_2 = -k_p x_1 - k_i \int_0^t x_1(\tau)d\tau - k_d \dot{x}_1 \underbrace{-x_1^3 - x_2^3 + x_1^3 + x_2^3}_{\text{Cancellation}}$$
$$= -k_p x_1 - k_i \int_0^t x_1(\tau)d\tau - k_d \dot{x}_1 \tag{8}$$

which can be expressed compactly as

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \quad \text{with} \quad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k_i & -k_p & -k_d \end{bmatrix}. \tag{9}$$

By pole placement, if we wish to place the three poles of the closed-loop system at $s_1 = s_2 = s_3 = -a$, here $a > 0$ is a constant representing the magnitude of the pole, then the three parameters k_p, k_i, and k_d should be chosen as $3a$, $3a^2$, and a^3, respectively, which ensures the exponential stability of the closed-loop system.

Remark 2: Through theoretical analysis, we identify three fundamental distinctions between linear and nonlinear PID control. 1) Nonlinearity compensation: Linear PID controllers with constant gains cannot counteract inherent nonlinear terms (e.g., x_1^3 and x_2^3) in the system dynamics, rendering these nonlinearities a persistent source of system destabilization. In contrast, the proposed nonlinear PID controller actively compensates for such terms through state-aware components in the control law. 2) Gain regulation mechanism: Linear PID employs fixed gains (k_p, k_i, and k_d) regardless of operating conditions, whereas nonlinear PID features state-aware gains ($K_P(x_1, k_p) = k_p + x_1^2$, $K_D(x_2, k_d) = k_d + \dot{x}_1^2$) that vary with system states, providing dynamic nonlinearity compensation. 3) Stability guarantees: Linear PID stability relies on local linearization, often failing to ensure global stability due to unaccounted nonlinear effects. The nonlinear PID controller achieves global exponential stability for arbitrary initial conditions through proper pole placement.

This motivating example demonstrates that, for nonlinear systems, linear PID control using fixed gains is bound to fail, regardless of how the gains are tuned. By contrast, using nonlinear state-dependent (error-coupled) PID gains yields excellent control performance, justifying the need for nonlinear PID control [14]. This need becomes even more critical in the presence of additional uncertainties (e.g., external disturbances), which demand self-adjusting nonlinear gains to achieve both robustness and adaptivity, as explored in this work.

IV. Main Results

Motivated by the above observation, we explore nonlinear PID strategies to address the aforementioned limitations of conventional linear PID control: 1) its inability to handle nonlinear dynamics effectively and 2) integral windup, a well-known drawback in traditional PID methods.

A. Key Idea and Design Philosophy

We propose an error-aware gain regulation method, forming a nonlinear PID control framework. The philosophy behind fabricating error-driven nonlinear PID gains, along with the key ideas for integrating these gains into the PID controller, is detailed in the following sections.

1) PID Filter With Error-Aware Nonlinear Gains: First, we introduce a PID filter driven by nonlinear error-aware gains as follows:

$$S = K_P(z, k_p)z + K_I\left(\int_0^t z(\tau)d\tau, k_i\right)\int_0^t z(\tau)d\tau + K_D(\dot{z}, k_d)\dot{z} \tag{10}$$

where z is the computable general tracking error defined in (2). $K_P(z, k_p) = \text{diag}[K_{Pi}(z_i, k_{pi})] \in \mathbb{R}^{m\times m}$, $K_I(\int_0^t z(\tau)d\tau, k_i) = \text{diag}[K_{Ii}(\int_0^t z_i(\tau)d\tau, k_{ii})] \in \mathbb{R}^{m\times m}$, and $K_D(\dot{z}, k_d) = \text{diag}[K_{Di}(\dot{z}_i, k_{di})] \in \mathbb{R}^{m\times m}$ denote the proportional, integral, and derivative gain diagonal positive-definite matrix, respectively, and k_p, k_i, and k_d are user-designed parameter vectors of appropriate dimensions with $k_{pi} > 0$, $k_{ii} > 0$, and $k_{di} > 0$ being user-designed parameters.

As observed in (10), the nonlinear PID gains—$K_P(z, k_p)$, $K_I(\int_0^t z(\tau)d\tau, k_i)$, and $K_D(\dot{z}, k_d)$—are correlated with their respective error signals: z, $\int_0^t z(\tau)d\tau$, and $\dot{z}$. These gains are self-adjusted in response to variations in the error signals, making them "aware" of the error evolution. This self-adjustment mechanism, where gains change in accordance with error variations, is why the approach is termed an *error-aware* (or *error-powered*) gain-based PID control method.

Notably, while each gain can be linked with the error $z(t)$, $\int_0^t z(\tau)d\tau$, and $\dot{z}(t)$, simultaneously, here we deliberately employ a streamlined design architecture such that each gain is restricted to depend solely on its corresponding error component, as defined in (10). This intentional simplification achieves two critical objectives: 1) it preserves structural transparency in the controller design, ensuring intuitive parameter tuning and physical interpretability and 2) it establishes a foundation for rigorous analytical tractability, enabling systematic stability analysis and performance guarantees in subsequent sections. By balancing generality with methodological pragmatism, this approach retains the nonlinear adaptability of error-aware gains while circumventing the mathematical complexities inherent in fully coupled gain dependencies.

Designing the error-aware proportional gain $K_{Pi}(z_i, k_{pi})$: The proportional gain $K_{Pi}(z_i, k_{pi})$ is designed based on the following principles:

1) $K_{Pi}(0, k_{pi}) = k_{pi}$;
2) $K_{Pi}(z_i, k_{pi})$ is bounded if z_i is bounded with $K_{Pi}(z_i, k_{pi}) \geq k_{pi} > 0$; and
3) $K_{Pi}(z_i, k_{pi})$ is monotonically increasing with respect to $|z_i|$.

Remark 3: With the error-aware proportional gain, the following notable features are achieved.

1) *High gain for large errors:* When the magnitude of error z_i is large, $K_{Pi}(z_i, k_{pi})$ increases with $|z_i|$, enhancing the control effort, accelerating the system's response speed.
2) *Low gain for small errors:* As the magnitude of error z_i decreases, $K_{Pi}(z_i, k_{pi})$ decreases accordingly, avoiding overshoot and oscillation caused by excessive gain and improving steady-state accuracy.

For example, the following two choices for $K_{Pi}(z_i, k_{pi})$:

$$K_{Pi}(z_i, k_{pi}) = k_{pi} + k_{pi} z_i^2 \tag{11}$$

$$K_{Pi}(z_i, k_{pi}) = k_{pi} \exp(z_i^2) \tag{12}$$

as illustrated in Fig. 1 , which clearly shows the monotonic increase of $K_{Pi}(z_i, k_{pi})$ with respect to $|z_i|$, in line with the design principles and characteristics discussed above.

Designing the error-aware integral gain $K_{Ii}(\int_0^t z_i(\tau)d\tau, k_{ii})$: The error-aware integral gain $K_{Ii}(\int_0^t z_i(\tau)d\tau, k_{ii})$ is chosen in light of the following rules:

1) $K_{Ii}(0, k_{ii}) = k_{ii}$;
2) $K_{Ii}(\int_0^t z_i(\tau)d\tau, k_{ii}) \cdot \int_0^t z_i(\tau)d\tau$ is bounded for $\forall t \geq 0$;

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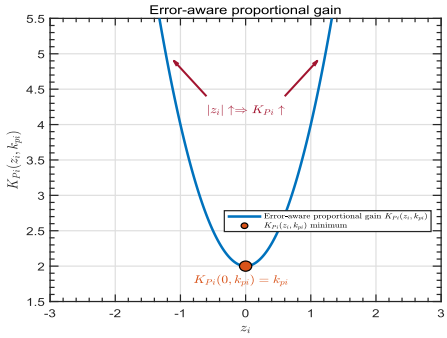


Fig. 1. Error-aware proportional gain $K_{Pi}(\cdot)$ as in (11).

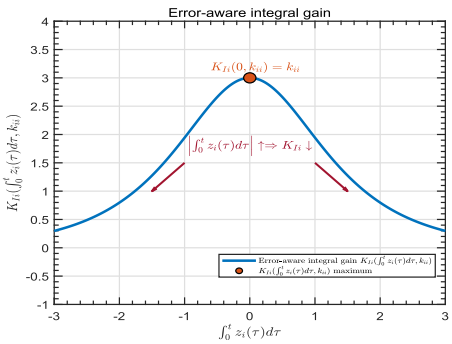


Fig. 2. Error-aware integral gain $K_{Ii}(\cdot)$ as in (13).

3) $K_{Ii}(\int_0^t z_i(\tau)d\tau, k_{ii}) \in (0, k_{ii}]$ with $k_{ii} > 0$; and
4) $K_{Ii}(\int_0^t z_i(\tau)d\tau, k_{ii})$ is monotonically decreasing with respect to $|\int_0^t z_i(\tau)d\tau|$.

Remark 4: Using the error-aware integral gain as designed offers several benefits.

1) *Anti-Windup Property:* The fact that $K_{Ii}\left(\int_0^t z_i(\tau)d\tau, k_{ii}\right)$ decreases as the accumulated error magnitude $\left|\int_0^t z_i(\tau)d\tau\right|$ increases}, automatically limits integral windup; and
2) *Steady-State Regulation:* It is seen from Fig. 2 that K_{Ii} tends to K_{ii} as $\int_0^t z_i(\tau)d\tau$ tends to zero, indicating that high integral gain K_I sustains as error accumulation diminishes, achieving static error removal without excessive control actions.

Some possible choices are

$$K_{Ii}\left(\int_0^t z_i(\tau)d\tau, k_{ii}\right) = k_{ii}\text{sech}\left(\int_0^t z_i(\tau)d\tau\right) \tag{13}$$

$$K_{Ii}\left(\int_0^t z_i(\tau)d\tau, k_{ii}\right) = k_{ii}\exp\left(-\left(\int_0^t z_i(\tau)d\tau\right)^2\right) \tag{14}$$

from which it is seen that $K_{Ii}\left(\int_0^t z_i(\tau)d\tau, k_{ii}\right) \cdot \int_0^t z_i(\tau)d\tau$ remains bounded even if the accumulated error $\left(\int_0^t z_i(\tau)d\tau\right)$ is large, while for small accumulated error $|\int_0^t z_i(\tau)d\tau| < \delta_i$ with δ_i being small, K_{Ii} tends to k_{ii}, namely, a higher integral gain is maintained within steady-state regions to enhance the integral action, effectively eliminating static errors. In particular, $K_{Ii}(0,) = k_{ii}$, as shown in Fig. 2. Interestingly, with such setting, integral action $K_{Ii}\left(\int_0^t z_i(\tau)d\tau, k_{ii}\right) \cdot \int_0^t z_i(\tau)d\tau$ is always bounded, thereby anti-windup is achieved elegantly with the proposed method. Note that the adjustment process for $(K_{Ii}(\cdot))$ is smooth and is executed seamlessly, ensuring that there are no sudden changes in the control signal.

Remark 5: It is important to note that, to effectively mitigate integral windup, the integral gain $K_{Ii}(\cdot)$ should not remain constant. This is because, as the system error continuously accumulated (i.e., $\int_0^t z_i(\tau)d\tau$ becomes large), it can easily lead to integral saturation, potentially causing system instability. To address this issue, this article proposes for the first time an error-based gain adjustment mechanism that enables $K_{Ii}(\cdot)$ to be continuously adjusted according to the variation of $\int_0^t z_i(\tau)d\tau$. This approach ensures the effective utilization of integral control while significantly suppressing saturation phenomena caused by the integral term, clearly demonstrating the novelty of the design philosophy based on smooth error-aware gain adjustment. This constitutes another key motivation for adopting the nonlinear PID control structure in this study.

Designing the error-aware derivative gain $K_{Di}(\dot{z}_i, k_{di})$: The error-aware derivative gain $K_{Di}(\dot{z}_i, k_{di})$ is designed to exhibit the following traits:

1) $K_{Di}(0, k_{di}) = k_{di}$;
2) $K_{Di}(\dot{z}_i, k_{di})$ is bounded if $\dot{z}_i$ is bounded with $K_{Di}(\dot{z}_i, k_{di}) \geq k_{di} > 0$; and
3) $K_{Di}(\dot{z}_i, k_{di})$ is monotonically increasing with respect to $|\dot{z}_i|$

Remark 6: The error-driven derivative gain is exemplified by the following functions:

$$K_{Di}(\dot{z}_i, k_{di}) = k_{di} + k_{di}\dot{z}_i^2 \tag{15}$$

$$K_{Di}(\dot{z}_i, k_{di}) = k_{di}\ln(\exp(1) + \dot{z}_i^2) \tag{16}$$

with behaviors similar to that as illustrated in Fig. 1. Basically, with the proposed setting for K_Di, when the magnitude of the error rate $(\dot{z}_i)$ is small, a lower gain $K_{Di}(\dot{z}_i, k_{di})$ is generated. This is well aligned with the practical situation because in this case it does not require significant driving actions. However, if $|\dot{z}_i|$ is large, a large $K_{Di}(\dot{z}_i, k_{di})$ is necessary so that substantial corrective control actions are produced accordingly .

To simplify our presentation in the remaining part of this article, we will abbreviate $K_P(\dot{z}, k_p)$, $K_I(\int_0^t z(\tau)d\tau, k_i)$ and $K_D(\dot{z}, k_d)$ as $K_P(\cdot)$, $K_I(\cdot)$ and $K_I(\cdot)$, respectively.

2) Useful Lemmas: The design philosophy is to make $S(t)$ in (10) converge to zero asymptotically, then by Lemma 2, it is established that $z(t)$ converges to zero asymptotically, here Lemma 1 is used to help deriving Lemma 2.

Lemma 1: For each $i = 1, \ldots, m$, let $\xi_i(t) = \int_0^t z_i(\tau)\,d\tau$, and $\frac{K_{Ii}(\theta_i, k_{ii})}{K_{Di}(f_i(\theta_i), k_{di})} > 0$ for all $\theta_i, f_i(\theta_i) \in \mathbb{R}$. Then, the integral

$$\mathbf{V}_1 = \int_0^{\xi_i(t)} \frac{K_{Ii}(\theta_i, k_{ii})}{K_{Di}(f_i(\theta_i), k_{di})}\theta_i\,d\theta_i \geq 0$$

for all $\xi_i(t) \in \mathbb{R}$.

Lemma 2: Let $S(t)$ be generated by (10), if $S(t) \to \mathbf{0}_m$ as $t \to \infty$, then $z(t) \to \mathbf{0}_m$ as $t \to \infty$.

Proof: See the Appendix for the detailed proofs of Lemmas 1 and 2. ■

Lemma 3 (See [19]): Let the generalized error z be defined as in (2). If $z(t) \to \mathbf{0}_m$ as $t \to \infty$, then $e^{(k)}(t) \to \mathbf{0}_m$ as $t \to \infty$ $(k = 0, \ldots, n-2)$.

B. Nonlinear PID Control for Nonlinear Systems

We are now ready to develop the nonlinear PID control with the three error-aware gains defined previously. To this end, we generate the composite signal S via (10), upon which we construct the following adaptive nonlinear PID-like control strategy:

$$u = -(k + k(t))\Lambda S \tag{17}$$

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where $k > 0$, $k(t) = \frac{\hat{b}\psi^2(\cdot)}{\sqrt{\psi^2(\cdot)\|S\|^2+\varepsilon^2(t)}}$, $\Lambda = \frac{\partial K_D(\cdot)}{\partial \dot{z}}\dot{z} + K_D(\cdot)$ is a positive-definite matrix, and the adaptive parameter

$$\dot{\hat{b}} = \frac{\gamma\psi^2(\cdot)\|S\|^2}{\sqrt{\psi^2(\cdot)\|S\|^2 + \varepsilon^2(t)}}, \quad \hat{b}(0) \geq 0 \tag{18}$$

where $\hat{b}$ is the estimate of the virtual parameter b [a virtual parameter to be defined in (26)], $\gamma > 0$ is a user designed constant, $\psi(\cdot)$ is a nonlinear function as defined by (27), and $\varepsilon(t)$ is a positive scalar function chosen to satisfy $\int_0^t \varepsilon(\tau)d\tau < \bar{\varepsilon} < \infty$ for $\forall t \geq 0$, with $\bar{\varepsilon}$ being a positive constant.

Theorem 1: Consider system (1), satisfying Assumptions 1–3. By applying the control scheme (17) with adaptive law (18), asymptotic tracking is achieved, i.e., $e(t)$ tends to zero as time goes to infinity.

Proof: See the Appendix for detail proof. ■

Remark 7: The proposed nonlinear PID controller distinguishes itself from conventional PID methods through three key features.

1) *Adaptive nonlinear structure:* While retaining the simplicity of classical PID control, the design employs nonlinear, error-aware gains for proportional, integral, and derivative actions. These gains self-adjust continuously based on: ① instantaneous tracking error magnitude, ② temporal accumulation of error, and ③ rate of error evolution. This mechanism actively compensates for system nonlinearities and disturbances, significantly improving control accuracy and robustness, thereby overcoming the limitations of fixed gains in traditional PID methods.
2) *Inherent antiwindup design:* As the integral gains formulated as $K_{Ii}(\cdot) = k_{ii}\text{sech}(\int_0^t z_i(\tau)d\tau)/k_{ii}\exp(-(\int_0^t z_i(\tau)d\tau)^2)$, such structure can be ensured smooth attenuation of integral action during large error accumulation, preventing saturation while preserving steady-state precision through bounded integration.
3) *Practical nonlinear control framework:* By synergizing adaptive nonlinear compensation and antiwindup mechanism within a PID architecture, the controller provides an appealing solution for tracking control of nonlinear systems with uncertainties balancing theoretical rigor and implementation feasibility.

Remark 8: A recent work [15] explored nonlinear PID control for second-order SISO normal form systems. Our proposed nonlinear PID control method is designed for MIMO nth-order systems, requiring no a priori knowledge of control gains, imposing no stringent conditions, such as vanishing second-order partial derivatives or Lipschitz continuity on $f(x_1, x_2)$, inherently addressing integral windup, and supporting time-varying reference tracking rather than set-point regulation.

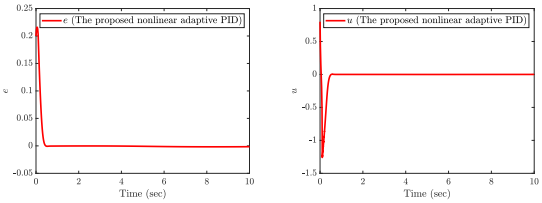
Remark 9: It is interesting to observe that while stability analysis makes use of various parameters (λ, b^*, and $\bar{d}$), these parameters are not incorporated into the final control algorithms. As a result, there is no need for analytically estimating or deriving these parameters in control design or implementation. In designing and implementing the proposed strategy, one only needs to choose $k > 0$, $k_{pi} > 0$, $k_{ii} > 0$, and $k_{di} > 0$ quite arbitrarily. This characteristic renders the proposed control scheme structurally simple and computationally effective.

V. Simulation Verification

To comprehensively and systematically verify the effectiveness and performance of our proposed adaptive nonlinear PID control strategy, we designed two simulation experiments to verify it.

Example 1. Nonlinear systems with unknown nonlinearity and disturbances: Reconsider the system (4) with external disturbances

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = u + f(x_1, x_2) + d(t) \end{cases} \tag{19}$$



(a) (b)

Fig. 3. Tracking error (e) and the control input (u). (a) Tracking error (e) (rad). (b) Control input (u) (Nm).

where $x_1, x_2 \in \mathbb{R}$ are the system states, $u \in \mathbb{R}$ is control input, $f(x) = x_1^3 + x_2^3$ is a unknown nonlinear function, and $d(t)$ is external disturbance (in the simulation, $d(t)$ is set as $0.5\sin(0.5t)$).

In the simulation, the tracking error is defined as $z = e = x_1 - y^*$, where y^* represents the desired signal, which is set to $y^* = 0.5\cos(0.5t)$. Meanwhile, the initial conditions of the system are set to $x_1(0) = 0.7$ and $x_2(0) = 0.8$, and the control parameters are configured as $k = 1$, $\gamma = 0.5$, and $\varepsilon(t) = 0.1\exp(-0.1t)$. In addition, the initial value of the estimated parameter $\hat{b}$ is set to $\hat{b}(0) = 0$. Finally, the nonlinear PID gains are designed according to (11), (13), and (15), with the parameters set to $k_p = 15$, $k_i = 1$, and $k_d = 0.1$.

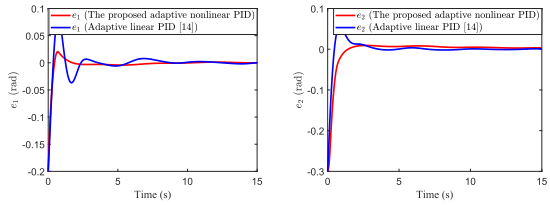
The simulation results are presented in Fig. 3, from which it is evident that the proposed nonlinear adaptive PID controller achieves remarkable outcomes: 1) the tracking error e rapidly converges to zero and 2) the control input remains continuous and bounded. These results verify the effectiveness of the proposed adaptive nonlinear PID control in addressing nonlinearity $f(x_1, x_2)$ and external disturbances $d(t)$, further highlighting the elegance and superiority of the proposed controller.

Example 2. Application to Robotic System: We now examine the applicability of the proposed nonlinear PID design approach for the well-known robotic systems. Consider a two joint rigid-link robotic manipulator with the following joint-space dynamics; for specific details, see [18]

$$\mathcal{M}(q, \ell)\ddot{q} + \mathcal{W}_c(q, \dot{q}, \ell)\dot{q} + \mathcal{G}_g(q, \ell) + \tau_d = u. \tag{20}$$

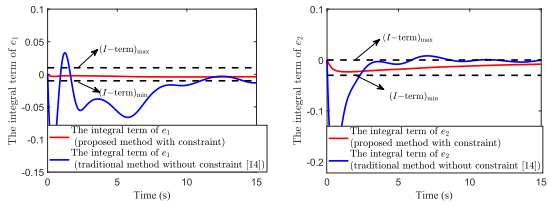
For the purpose of simulation, we define the tracking output error as $z = e = q - y^*$, where the desired signal $y^* = [0.3\cos(0.5t), 0.4\cos(0.5t)]^\top$ (rad), and the initial conditions are $y^*(0) = [0, 0]^\top$ (rad), $q(0) = [0.1, 0.1]^\top$ (rad), $\dot{q}(0) = [0.8, 0.1]^\top$(rad/s), $k = 1$, $\gamma = 0.5$, $\varepsilon(t) = 0.1\exp(-0.1t)$, and $\hat{b}(0) = 0$. The nonlinear PID is given as in (17) with $S = K_P(z, k_p)z + K_I(\int_0^t z(\tau)d\tau, k_i)\int_0^t z(\tau)d\tau + K_D(\dot{z}, k_d)\dot{z}$, in which the gains are selected as those previously defined in (11), (13), and (15), with the parameters set accordingly and quite arbitrarily as $k_{pi} = 2.5$, $k_i = 1$, and $k_{di} = 2.1$. It is worth emphasizing that although the nonlinear PID controller proposed in literature [15] solves the regulation problem for second-order SISO nonlinear systems, it is inapplicable to robotic system (20) as discussed in Remark 8. In particular, the system here is MIMO and the control gain is no longer unit but a unknown and time-varying matrix, yet the nonlinear function does not satisfy the restrictive conditions as imposed in [15], whereas our method is readily applicable to such a typical nonlinear MIMO system. To validate the advantages of our proposed algorithm, the adaptive linear PID controller with time-varying gains in [13] is also applied for comparison. The results are compared in Figs. 4–7. As shown in the error response curves, the nonlinear control approach drives the system error to zero within a shorter time, with noticeably smaller overshoot and reduced oscillation amplitude in

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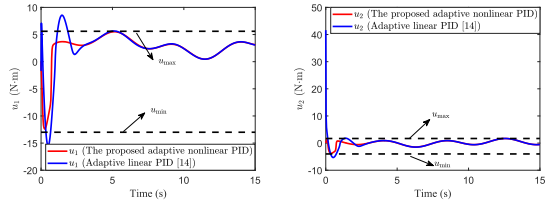
(a) (b)

Fig. 4. Comparison of the tracking error (e_i) $((i = 1, 2))$. (a) Tracking error (e_1) (rad). (b) Tracking error (e_2) (rad).



(a) (b)

Fig. 5. Comparison of the integral terms' dynamic responses. (a) Integral term of (e_1). (b) Integral term of (e_2).



(a) (b)

Fig. 6. Comparison of the control action (u_i) $((i = 1, 2))$. (a) Control action (u_1) (N/m). (b) Control action (u_2) (N/m).

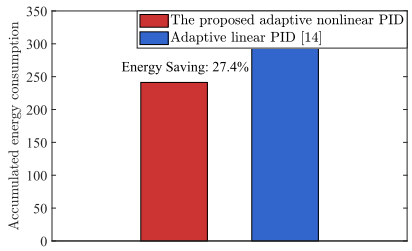


Fig. 7. Performance index $J_u = \int_0^t \|u(\tau)\|^2 d\tau$ comparison.

Fig. 4. Fig. 5 clearly indicates the distinct behaviors of the two algorithms in dealing with saturation: the integral term of the conventional method significantly exceeds its effective range due to uncontrolled accumulation, whereas the dynamic behavior of the integral term in our proposed algorithm is well constrained within the valid range, thereby preventing the occurrence of windup. As shown in Fig. 6, the control inputs generated by the proposed nonlinear PID controller are smoother and of lower amplitude, *while* strictly adhering to the prescribed bounds $(u_{\min}, u_{\max})$; in contrast, the method from [14] violates the physical limits of the actuator. This demonstrates that the proposed scheme effectively achieves the control objectives while avoiding aggressive input changes and actuator saturation. From Fig. 7, in terms of accumulated energy consumption, the nonlinear PID controller exhibits a higher level of energy efficiency. As illustrated in the energy comparison bar chart, the accumulated energy saving (operating 15 s) as compared with adaptive linear PID control is over 27%.

VI. Conclusion

Traditional PID control exhibits significant limitations when dealing with unknown complex nonlinear systems. This article proposes a novel PID tracking control method incorporating nonlinear gains and parameter self-adjustment for a class of unknown coupled nonlinear systems. By systematically utilizing current error, past error, and predictive error to construct the corresponding nonlinear PID gains analytically, while introducing single-parameter adaptive adjustment, the method achieves asymptotic tracking. The proposed approach maintains the intuitive structure and implementation simplicity of conventional PID control while endowing the controller with inherent nonlinear compensation capabilities, along with both adaptive and robust characteristics. Furthermore, the method features antiwindup capability and operates completely autonomously without requiring manual intervention. Future research may focus on extending this method to more general nonlinear systems and exploring additional effective nonlinear gain functions to further enhance the controller's adaptability and robustness in complex systems.

Appendix

Proof of Lemma 1: We only need to analyze two cases based on the sign of $\xi_i(t)$.

1) $\xi_i(t) \geq 0$, the integration interval is $[0, \xi_i(t)]$. Since θ_i in this interval is positive and using the fact that $\frac{K_{Ii}}{K_{Di}} > 0$, it follows that $\mathbf{V}_1$ is nonnegative.
2) $\xi_i(t) < 0$, then in the interval $[\xi_i(t), 0)$, $\theta_i \leq 0$, making the integrand $\frac{K_{Ii}}{K_{Di}}\theta_i \leq 0$. It follows that:

$$\int_0^{\xi_i(t)} \frac{K_{Ii}}{K_{Di}}\theta_i \, d\theta_i = -\int_{\xi_i(t)}^0 \frac{K_{Ii}}{K_{Di}}\theta_i \, d\theta_i \geq 0.$$

Thus, $\mathbf{V}_1$ is nonnegative for all $\xi_i(t), \theta_i, f_i(\theta_i) \in \mathbb{R}$.

Proof of Lemma 2: We construct the following Lyapunov candidate:

$$\mathbf{V}_0 = \frac{1}{2}z^T z + \sum_{i=1}^m \int_0^{\int_0^t z_i(\tau)\, d\tau} \frac{K_{Ii}(\theta_i, k_{ii})}{K_{Di}(f_i(\theta_i), k_{di})}\theta_i \, d\theta_i \tag{21}$$

where in our design, we set $f_i(\int_0^t z_i(\tau)d\tau) = \dot{z}_i$ to facilitate the design. By Lemma 1, the second term is nonnegative, ensuring that $\mathbf{V}_0$ is nonnegative. □

Taking the time derivative of (21) and using (10) gives

$$\dot{\mathbf{V}}_0 \leq \sum_{i=1}^m \frac{1}{K_{Di}(\cdot)}|z_i||S_i| - \sum_{i=1}^m \frac{K_{Pi}(\cdot)}{K_{Di}(\cdot)}z_i^2. \tag{22}$$

Now, we consider the following two cases to show that z tends to zero as time tends to infinity.

1) *Case 1:* When $\sum_{i=1}^m \frac{1}{K_{Di}(\cdot)}|z_i||S_i| - \sum_{i=1}^m \frac{K_{Pi}(\cdot)}{K_{Di}(\cdot)}z_i^2 \leq 0$. Under this case, since $\mathbf{V}_0 \geq 0$ and $\dot{\mathbf{V}}_0 \leq 0$, $\mathbf{V}_0$ is nonincreasing. By LaSalle's invariance principle, $\mathbf{V}_0$ converges to the largest invariant set where $\dot{\mathbf{V}}_0 = 0$, i.e., $\sum_{i=1}^m \frac{1}{K_{Di}(\cdot)}|z_i||S_i| - \sum_{i=1}^m \frac{K_{Pi}(\cdot)}{K_{Di}(\cdot)}z_i^2 = 0$, implying $\sum_{i=1}^m K_{Pi}(\cdot)|z_i| = \sum_{i=1}^m |S_i|$ and $K_{Pi}(\cdot) \geq k_{pi} > 0$. Note that $\lim_{t\to\infty} S_i(t) = 0$, it follows that $\lim_{t\to\infty}|z_i(t)| = 0$ for all i. Thus, $\lim_{t\to\infty} z(t) = \mathbf{0}_m$.
2) *Case 2:* When $\sum_{i=1}^m \frac{1}{K_{Di}(\cdot)}|z_i||S_i| - \sum_{i=1}^m \frac{K_{Pi}(\cdot)}{K_{Di}(\cdot)}z_i^2 \geq 0$, then $\sum_{i=1}^m \frac{K_{Pi}(\cdot)}{K_{Di}(\cdot)}z_i^2 \leq \sum_{i=1}^m \frac{1}{K_{Di}(\cdot)}|z_i||S_i|$. Then, we have: $\sum_{i=1}^m K_{Pi}(\cdot)|z_i| \leq \sum_{i=1}^m |S_i|$ with $K_{Pi}(\cdot) \geq k_{pi} > 0$. Since $\lim_{t\to\infty} S_i(t) = 0$, it follows that $\lim_{t\to\infty}|z_i(t)| = 0$. Hence, $\lim_{t\to\infty} z(t) = \mathbf{0}_m$.

Thus, it is proved that in both cases, $\lim_{t\to\infty} z(t) = \mathbf{0}_m$ as $\lim_{t\to\infty} S(t) = \mathbf{0}_m$. □

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Proof of Theorem 1: Using (3) and (10), we derive the time derivative of S as

$$\dot{S} = \Lambda B(x,t)u + \Psi(\cdot) \tag{23}$$

where $\Lambda = \frac{\partial K_D(\cdot)}{\partial \dot{z}}\dot{z} + K_D(\cdot)$ is a user-designed positive - matrix and $\Psi(\cdot) = \Lambda L(\cdot) + \frac{d(K_P(\cdot)z)}{dt} + \frac{d(K_I(\cdot)\int_0^t z(\tau)d\tau)}{dt}$.

Note that the resultant control gain matrix $\Lambda B(x,t)$ in (23) is unknown and time-varying yet not even symmetric. In order to carry out Lyapunov-based stability analysis, we introduce the matrix $B_p = \frac{\Lambda(B+B^T)\Lambda}{2}$ and $\tilde{B}_s = \frac{\Lambda(B-B^T)\Lambda}{2}$ such that $\Lambda B\Lambda = B_{p_{\text{positive definite}}} + \tilde{B}_{s\,\text{skew symmetric}}$. With Assumption 1, B_p is positive definite and $\tilde{B}_s$ is skew symmetric; thus, there exists an unknown constant λ such that $0 < \lambda \leq \lambda_{\min}(B_p)$ and $S^T \tilde{B}_s S = 0$.

Choosing the Lyapunov function candidate as

$$V = \frac{1}{2}S^T S + \frac{1}{2\gamma\lambda}\tilde{b}^2 \tag{24}$$

where the estimate error is defined as $\tilde{b} = b - \lambda\hat{b}$ with b being a virtual parameter defined in (26). Taking the time derivative of (24) along (23) gives

$$\begin{aligned}\dot{V} &= S^T\Lambda B u + S^T\Psi(\cdot) - \frac{1}{\gamma}\tilde{b}\dot{\hat{b}} \\ &\leq -k\lambda\|S\|^2 - \lambda\frac{\hat{b}\psi^2(\cdot)\|S\|^2}{\sqrt{\psi^2(\cdot)\|S\|^2 + \varepsilon^2(t)}} + S^T\Psi(\cdot) - \frac{1}{\gamma}\tilde{b}\dot{\hat{b}}. \end{aligned}\tag{25}$$

Similarly, using Assumption 2 for $\|F(x,t)\|$, it follows that $\|F(x,t)\| \leq b^*\psi^*(x)$, then we have $\|\Psi(\cdot)\| \leq b\psi(\cdot)$ where

$$b = \max\{1, b^*, \bar{d}\} > 0 \tag{26}$$

and

$$\begin{aligned}\psi(\cdot) = &\left\| \frac{\partial(K_P(\cdot)z)}{\partial t} + \frac{\partial\left(K_I(\cdot)\int_0^t z(\tau)d\tau\right)}{\partial t} \right\| \\ &+ \|\Lambda\|\left(\psi^* + 1 + \| - y^{*(n)} + \iota_{n-3}e^{(n-1)} + \cdots + \iota_0\ddot{e}\|\right).\end{aligned}\tag{27}$$

Therefore, it follows that:

$$S^T\Psi(\cdot) \leq b\psi(\cdot)\|S\| \leq b\varepsilon(t) + \frac{b\psi^2(\cdot)\|S\|^2}{\sqrt{\psi^2(\cdot)\|S\|^2 + \varepsilon^2(t)}}. \tag{28}$$

By (28), using updating law $\dot{\hat{b}}$ in (18), it is not difficult to show from (25) that

$$\dot{V} \leq -\sigma\|S\|^2 + b\varepsilon(t) \tag{29}$$

where $\sigma = \lambda k > 0$. By integrating (29) over $[0,t)$, we get

$$V(t) + \sigma\int_0^t \|S(\tau)\|^2 d\tau \leq V(0) + b\bar{\varepsilon}. \tag{30}$$

From (30), it is evident that both $V(t)$ and $\int_0^t \|S(\tau)\|^2 d\tau$ are bounded above by $V(0) + b\bar{\varepsilon}$. Consequently, we can deduce that $V(t) \in L_\infty$ and $\int_0^t \|S(\tau)\|^2 d\tau \in L_\infty$. This implies that S belongs to L_2. Furthermore, by (30), it follows that $\|S\| \in L_\infty$ and $\tilde{b} = b - \lambda\hat{b} \in L_\infty$, which indicates that $\hat{b}$ is bounded. According to Lemma 2, the boundedness of $\|S\|$ ensures that both $\|z\|$ is bounded, then by (10), we can also conclude that $\|\dot{z}\|$ is bounded. This, in turn, implies that $e^{(k)}$ $(k = 0, \ldots, n-1)$ are bounded. Since both $e^{(k)}$ and $y^{*(k)}$ are bounded, it follows that $x_1^{(k)}$ is bounded, thereby ensuring that $\psi^*(\cdot)$ and $\psi(\cdot)$ are in L_∞. Subsequently, it can be readily shown that $k(t)$ is bounded for $t \in [0,\infty)$. Therefore, from (17), (18), and (23), we conclude that $\|u\|$, $\dot{\hat{b}}$, and $\|\dot{S}\|$ are bounded. This implies that all signals in the closed-loop system are bounded and continuous for all $t \in [0,\infty)$. Besides, given that $\|S\| \in L_\infty \cap L_2$ and $\|\dot{S}\| \in L_\infty$, by the Barbalat lemma, it can be established that $\lim_{t\to\infty}\|S(t)\| = 0$. According to Lemma 1, it follows that $\lim_{t\to\infty}\|z(t)\| = 0$. Then, by Lemma 3, we have $\lim_{t\to\infty} e = \mathbf{0}_m$. Hence, asymptotic tracking is achieved under the controller (17). □

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